

A SCENARIO FOR A LARGE NUMBER OF LOW-MASS BLACK HOLES IN THE GALAXY

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ABSTRACT

We point out that stars within a fairly large range of masses, roughly $18\text{--}30 M_{\odot}$, can both explode as supernovae (and give off neutrinos as observed in SN 1987A) and then go into black holes. The masses of the black holes so formed are only slightly above $1.5 M_{\odot}$, the maximum mass for compact cores, according to our arguments. These masses are substantially smaller than those of the best candidates for black holes observed thus far. We discuss the possibility that SN 1987A produced one of these small black holes.

We review the requirements that observed nucleosynthesis puts on theoretical nucleosynthesis. Especially the empirical value of $\Delta Y/\Delta Z = 4 \pm 1.3$ gives a sensitive determination of the mass at which nucleosynthesis must be cut off, by heavier stars collapsing into black holes without returning matter to the Galaxy, because otherwise heavy stars would produce chiefly metals and unduly lower the calculated value of $\Delta Y/\Delta Z$. We arrive at a value of $M_{\text{cutoff}} = 25 \pm 5 M_{\odot}$ for the mass at which nucleosynthesis must be cut off.

We show that the introduction of kaon condensation sufficiently softens the equation of state of dense matter, so that compact cores of mass greater than $M_{\text{max}} \simeq 1.5 M_{\odot}$ will not be stable. For the first $\gtrsim 12$ s, however, compact cores up to $\sim 1.84 M_{\odot}$ are stabilized; this gives the nucleosynthesis mass cutoff as $\sim 30 M_{\odot}$, at the upper end of our limit from nucleosynthetic demands. Our soft equation of state extends black hole production to stars of lower mass than previously estimated, and, therefore, increases the estimated number of black holes by an order of magnitude or more, to $\sim 10^9$ in the Galaxy.

Subject headings: black hole physics — Galaxy: stellar content — nuclear reactions, nucleosynthesis, abundances — stars: evolution

1. INTRODUCTION

It was suggested (Brown 1988) that the maximum neutron star mass could be $M_{\text{max}} \sim 1.5 M_{\odot}$, not much larger than the $1.442 \pm 0.03 M_{\odot}$ of the heavier pulsar (Taylor & Weisberg 1989) in PSR 1913+16. This suggestion was based on arguments that the high-density equation of state of dense matter is “softer” than generally thought. Recently (Brown 1992), with the implementation of the K^- condensation in the equation of state, this suggestion has become more compelling, as we shall discuss later. Rather than “neutron star” in what follows we shall use the term nucleon star, because with K^- condensation, nearly as many protons as neutrons are present.

Brown, Bruenn, & Wheeler (1992b) asked, “Is There a Black Hole in Supernova 1987A?” They suggested that the late time accretion or additional matter deposited by the reverse shock might send the core of the neutron star in SN 1987A over the limit of $\sim 1.50 M_{\odot}$ into a black hole after the nucleosynthesis and 12 s neutrino emission was completed. An alternative discussion of the late time accretion has since been given by Bethe (1993), which makes this scenario more definite.

For some time Twarog & Wheeler (1982, 1987) and Wheeler, Sneden, & Truran (1989) have argued that heavy stars at the end of their life must disappear from the Galaxy without ejecting their matter in a supernova explosion, to avoid overproduction of oxygen. They call this the “death function” and propose that masses above about $24\text{--}28 M_{\odot}$ disappear and presumably turn into black holes. These arguments were made more definite by Maeder (1992), who showed that stellar yields integrated over the mass spectrum are able to reproduce the

observed $\Delta Y/\Delta Z$ (the relative helium to metal enrichment) for low metallicities, only if there are black holes formed above $20\text{--}25 M_{\odot}$. Here ΔY is the increase of Y , the weight fraction of He, from the primordial value after the big bang, to its actual value, ΔZ , the same for “metals” where the primordial value is zero.

In this note we wish to make the scenario of black hole formation more definite, pointing out that stars within a fairly large range of masses can first explode as supernovae and give off neutrons, as observed in SN 1987A, and then go into black holes. The masses of the black holes so formed will not greatly exceed the $M_{\text{max}} \sim 1.5 M_{\odot}$ for nucleon stars, and, therefore, will be substantially smaller than those of the best candidates for black holes observed thus far.

Our paper will be organized in the following way. In § 2, we shall discuss the possibility that SN 1987A, as well as progenitors of similar mass, have gone into black holes. In § 3, we shall review the nucleosynthetic demands for a “death function” that sets in at relatively small stellar masses. In § 4, we discuss the soft equation of state of dense matter that results once a kaon condensate is introduced at several times the nuclear matter density. In § 5 we discuss the postshock mechanism for the formation of black holes. In § 6, we discuss the implications of our new scenario for the estimated number of black holes in the Galaxy.

2. NEUTRON STARS OR BLACK HOLES?

Helfand & Becker (1984), from radio, optical, X-ray, and γ -ray surveys of supernova remnants, came to the startling conclusion that nearly half of the supernovae in the Galaxy leave no observable compact remnant. In terms of historical supernovae in our galaxy, this was not necessarily surprising, since about half of them have been Type Ia, which are not expected to leave compact remnants. However, as discussed in

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detail by van den Bergh, McClure, & Evans (1987), the observed sample of supernovae in our Galaxy has been strongly biased against Type II supernovae (SN II's) because most of them are at low galactic latitude so their light is heavily absorbed on its way to us. These authors estimate from extragalactic studies (the supernovae rate in Shapley-Ames galaxies) that SN II's outnumber SN I's by a factor of about 6. (Type Ib supernovae, presumed to have heavy progenitors, are included with SN II in this estimate. We will return to a discussion of Type Ib supernovae at the end of § 5.) In any case, we can say that the SN II's outnumber the SN I's by a factor of several. Thus the Helfand-Becker conclusion can be understood only if almost half of the explosions from SN II's do not form compact remnants. We assume here the only viable alternative to be black holes. The conclusion—that roughly as many black holes as compact remnants are formed—is indeed startling, and we will spend the rest of this note to substantiate this possibility.

There is circumstantial evidence that black hole formation is associated with the supernova explosion Cas A. There is no sign of a nucleon star in Cas A as a thermal source of X-rays, a pulsar, or an X-ray synchrotron nebula. The fast-moving knots in Cas A show the Mg/CNO ratio to be down by a factor of 2 from solar while Si, S, Ar, and Ca are enhanced compared to CNO by a factor of a few (Chevalier & Kirshner 1979). This is roughly consistent with the ejecta expected from a $20 M_{\odot}$ star (Woosley & Weaver 1986). Thus Cas A is a good candidate for a star of about the same mass as the progenitor of SN 1987A that left a black hole remnant after exploding and producing nucleosynthesis. It should be noted that El Eid & Prantzos (1988) have suggested another scenario for Cas A. In it, the supernova results from disruption of a very massive star due to pair creation instability.

The light curve in SN 1987A is still being powered by radioactive decay and an additional source of luminosity greater than 8×10^{36} ergs s^{-1} is now ruled out by the bolometric light curve at day 1500 (Suntzeff et al. 1992). This lies far below early modeling of an interior pulsar which was assumed to deposit energy in the innermost layers at a constant rate of 8.5×10^{41} ergs s^{-1} (Shigeyama et al. 1987), and means that if a pulsar is present, its magnetic field is very weak and the star is rotating slowly. A more precise determination of the nature of the compact remnant must await the clearing of debris so that soft X-rays from the central star can be detected, but it seems reasonable to suggest that it is a black hole (Brown et al. 1992b).

Estimates of the compact core mass formed in the delayed explosion in SN 1987A (Mayle & Wilson 1988; Thielemann, Nomoto, & Hashimoto 1992) place it close to the $M_{\max} \sim 1.5 M_{\odot}$, so that the late time accretion, to be discussed in the § 5, could have pushed it over the limit.

All measured nucleon star masses, with the exception of Vela X-1 (Lamb 1991; Thorsett et al. 1993), are consistent with $M_{\max} \sim 1.5 M_{\odot}$, and the rather large 2σ error bar on Vela X-1 stretches below $1.6 M_{\odot}$. Thus, we believe the $M_{\max} \sim 1.5 M_{\odot}$ to be reasonable from the standpoint of nucleon star masses, although Vela X-1 is still above our limit. It should be noted that most equations of state in the literature have $M_{\max}$ between 2 and $3 M_{\odot}$, easily stabilizing Vela X-1.

3. NUCLEOSYNTHETIC DEMANDS ON THE STELLAR DEATH FUNCTION

As noted in § 1, Twarog & Wheeler (1982, 1987) have raised the problem that oxygen is overproduced unless there is a

cutoff at about $\sim 24\text{--}28 M_{\odot}$ on the main-sequence mass, above which the star does not contribute its content to the galaxy at the end of its life. This cutoff is called the “death function” (Wheeler et al. 1989). Recent work by Maeder (1992) has sharpened the nucleosynthetic requirements. Maeder focused on the ratio $\Delta Y/\Delta Z$ of the (relative) helium-to-metal enrichment. This ratio has been measured in extragalactic low-metallicity H II regions by Pagel et al. (1992), who give $\Delta Y/\Delta Z$ in the range of 3–6, with a preferred value between 4 and 5.

The quantity $\Delta Y/\Delta Z$ is sensitive to the maximum mass of a star that will give the products of the nucleosynthesis back to the galaxy. Stars of mass $M \lesssim 8 M_{\odot}$ shed their mass in the giant stage, mostly in the form of helium, before becoming white dwarfs. (They also shed hydrogen, but this is irrelevant for $\Delta Y/\Delta Z$.) Some helium, but mostly metals, are produced in heavier stars. The amount of produced oxygen, which comprises the main part of the metals, increases roughly linearly with the mass of the exploding star. If all the material in heavy stars, at the end of their lives, were to be given back to the Galaxy by supernova explosions, there would be too much metals, and $\Delta Y/\Delta Z$ would be between 1 and 2 (Maeder 1992). Therefore, many of the heavy stars must simply disappear; i.e., become black holes; we assume that all above a certain cutoff mass do so. The measured $\Delta Y/\Delta Z$ determines rather precisely where the cutoff should be made.

For low-metallicity regions, in which the influence of stellar winds is small, this ratio provides a stringent requirement on the cutoff. Maeder finds that stellar yields integrated over the mass spectrum are able to reproduce the observed $\Delta Y/\Delta Z$ if black holes are formed above $20\text{--}25 M_{\odot}$. (Note that this estimate refers to the lower limit on collapses which eject photosynthesized material.)

The initial work of Maeder suffered from normalization problems, resulting from his choice of a minimum mass $m_1 = 1 M_{\odot}$ in his equations. Most of the mass is, however, in stars with masses less than solar. Taking this into account (Maeder 1993) changes the absolute yields considerably. However, the relative enrichment $\Delta Y/\Delta Z$ is not affected by the normalization choice, so the main conclusion of Maeder's earlier paper—that the observed $\Delta Y/\Delta Z$ ratios in low-metallicity regions can be accounted for only with a mass cutoff of $M_{\text{cutoff}} \sim 20 M_{\odot}$ —is unchanged.

As we will point out, over a region of about $10 M_{\odot}$, collapsing stars eject photosynthesized material and then go into light-mass black holes. Thus our limit for black hole formation will lie somewhat below Maeder's and may encompass the $\sim 18 M_{\odot}$ estimated for the progenitor of SN 1987A.

A major uncertainty in Maeder's estimates is in the initial mass function (IMF). This is taken as a power law

$$\frac{dN}{dM} = M^{-(1+x)}, \quad (1)$$

i.e., the classical Salpeter mass function. In order to estimate uncertainties, calculations were also performed for $x = 1.70$. Maeder's results for $\Delta Y/\Delta Z$ show rather large differences between the two values, $x = 1.35$ and 1.70 , for the slope of the IMF. Of course, the cutoff is at a substantially higher mass in the case of the 1.70 slope, since the larger slope already decreases the frequency of heavy-mass stars so their contribution to nucleosynthesis does not have to be cutoff so early.

In a series of recent papers, Massey et al. (1989a); Massey, Parker, & Garmany (1989b); Massey & Thompson (1991), and

Parker et al. (1992) have measured the IMFs of stars up to $80 M_{\odot}$ in OB associations in the LMC, SMC, and in our Galaxy. These authors find that the IMF can be fitted, in the Salpeter form with slopes ranging from $x = 1.0 \pm 0.1$ to 1.8 ± 0.1 (Bildsten 1993). We find the total mass *not* ejected to be

$$\langle M \rangle = \int_{M_1}^{M_{\max}} M \left(\frac{dN}{dM} \right) dM = \frac{C}{x-1} \left[1 - \left(\frac{M_{\max}}{M_1} \right)^{1-x} \right], \quad (2)$$

where $M_{\max}$ is the maximum mass of a stable star, M_1 the cutoff mass, and $C = M_1^2 (dN/dM)_{M_1}$ is a normalization constant. Offhand, small x seem to play a disproportionately large role in equation (2). But equation (2) is easy to evaluate. For $M_1 = 25$ and $M_{\max} = 130$, we have

$$\begin{array}{ccc} x = 1.1 & 1.35 & 1.8 \\ \langle M \rangle / C = 1.50 & 1.13 & 0.91. \end{array}$$

Thus, the dependence on x is actually moderate, and Salpeter's $x = 1.35$ seems a good average. We do not believe that larger x 's need to be considered.

From an entire grid of evolved stars of 0.1 – $100 M_{\odot}$, Timmes, Woosley, & Weaver (1993) have calculated Galactic chemical evolution. They have extended their calculation (private communication) to compute $\Delta Y / \Delta Z$, confirming Maeder's results, but with a mass cut slightly higher by $\sim 5 M_{\odot}$ in order to fit the Pagel values. We tend to confirm their higher mass cut (see § 5).

We conclude from the above arguments that return of elements to the Galaxy from stellar collapse must be cut off at a mass

$$M_{\text{cutoff}} = 25 \pm 5 M_{\odot}. \quad (3)$$

It should be noted that Timmes et al. (1993) do not find the mass of their compact core to increase monotonically with stellar mass. Thus, for zero metallicity stars of mass 15, 20, 25, and $30 M_{\odot}$, the baryon number masses of the core are 1.62, 1.42, 1.90, and $1.87 M_{\odot}$, respectively. For solar metallicity, stars of mass, 15, 18, 20, 25, and $30 M_{\odot}$ have baryonic core masses of 1.57, 1.64, 1.54, 1.67, and $1.99 M_{\odot}$. Although the metallicity dependence of these masses is extremely interesting, we will neglect it in our present discussion. In order to convert these masses into gravitational masses, we use the binding energy given by Lattimer & Yahil (1989)

$$E = 0.075 M_{\odot} \left(\frac{M}{M_{\odot}} \right)^2, \quad (4)$$

which is a $0.17 M_{\odot}$ correction for $M = 1.5 M_{\odot}$, $0.3 M_{\odot}$ for $M = 2 M_{\odot}$. In our scenario, then the stars with baryon masses greater than $1.67 M_{\odot}$ will go ultimately into black holes. However, in § 5 we shall show that all of the stars of masses listed above will return matter to the galaxy, so that those that do go into black holes will go into light mass ones, $M_{\text{BH}} \gtrsim 1.5 M_{\odot}$.

Note that the $1.64 M_{\odot}$ for the $\sim 18 M_{\odot}$ progenitor of SN 1987A lies just below the limit of $1.67 M_{\odot}$ for stars to go into a black hole, although the late time accretion (see § 5) would send it into one. Since the mass of the progenitor was known only to within $\sim \pm 2 M_{\odot}$, it is obviously a delicate matter whether SN 1987A went into a black hole or not, but the possibility certainly exists.

4. K^- CONDENSATION AND THE EQUATION OF STATE OF DENSE MATTER

Central to our scenario is that the equation of state of dense matter, for $\rho \gtrsim 3\rho_0$, where ρ_0 is nuclear matter density, is very different from the conventional one. In the latter, most of the protons change into neutrons at the higher densities, because it is too expensive in terms of energy to keep many electrons. The ratio of electrons to nucleons at high densities is typically $Y_l \sim 0.1$ and the same number of protons as electrons will be present. Therefore, one talks about the compact remnants formed in stellar collapse as "neutron stars." In our new scenario, based on the developments of Brown et al. (1992a), it is energetically favorable for protons to remain at the higher densities $\rho \neq 3\rho_0$. Their charge is chiefly balanced by K^- mesons, which form a Bose condensate, although a few electrons may still be present. The energy of this Bose condensate is lowered because of strongly attractive interactions between nucleons and the K^- mesons. The net result of kaon condensation is that at the higher densities at which the stability of the nucleon stars is determined, nuclear matter, rather than neutron matter, is present. In other words, the compact objects are nuclear matter stars (or nucleon), rather than neutron stars. We will now briefly sketch the history of kaon condensation.

Some years ago, Kaplan & Nelson (1986) showed, starting from chiral Lagrangians, that kaons obtain an effective mass m_K^* which decreases in dense matter, going to zero at several times nuclear matter density. Explicit solution of the dense matter problem was given by Politzer & Wise (1991). Quantitative application to stellar collapse has been worked out by Brown et al. (1992a) and detailed calculations of the equation of state, with application to compact cores of collapsing stars, by Thorsson (1992) and Thorsson, Prakash, & Lattimer (1993). The calculations are somewhat involved, so we refer the reader to these papers in detail.

The chief result of the drop in m_K^* for us is that K^- mesons can replace electrons in carrying negative charge in the dense matter of collapsing stars. Thus, "kaon condensation" sets in when the *in-medium* kaon energy ω_{K^-} becomes equal to the electron chemical potential

$$\omega_{K^-} = \mu_- = \mu_n - \mu_p, \quad (5)$$

where the final equality is set by beta equilibrium. In addition to the kaon mass m_K^* being lowered, the K^- meson experiences an attractive vector mean field potential from the nucleon. This mean field originates from the same exchange of ρ - and ω -mesons that produce the $\Lambda(1405)$ as a bound state of $\bar{K}$ and nucleon (Siegel & Weise 1988; Müller-Groeling, Holinde, & Speth 1990). In somewhat conservative calculations, Thorsson (1992) and Brown et al. (1992a) find that kaon condensation sets in at a density $\rho_c \simeq 3\rho_0$, where $\rho_0 = 0.16 \text{ fm}^{-3}$ is nuclear matter density.

Of chief concern to us here is the fact that the effect of kaon condensation is to soften the equation of state of dense matter. Reasons for a low-compression modulus of nuclear matter were summarized in Brown (1988). The kaon condensate comes in at higher densities in order to keep the equation of state soft. This softness, as compared with neutron matter, chiefly results from the lower symmetry energy resulting from the nearly equal numbers of protons and neutrons in kaon-condensed matter.

Thorsson (1992) and Thorsson et al. (1993) have investigated the composition, structure, and evolution of nucleon stars with

kaon condensates. With equations of state that satisfy the empirical properties of nuclear matter, and have a compression modulus of $K_0 = 180$ MeV, at the lower end of the empirically quoted 210 ± 30 MeV (Blaizot, Gogny, & Grammaticos 1976), and with the kaon condensate, the authors find compact core masses of

$$1.42 M_\odot < M_{\max} < 1.45 M_\odot, \quad (6)$$

the precise value depending upon the assumed density dependence of the nuclear symmetry energy. These values are barely able to stabilize the heavier pulsar of mass 1.442 ± 0.03 in PSR 1913+16 (Taylor & Weisberg 1989). An increase of K_0 up to ~ 190 MeV would bring $M_{\max}$ up to our estimated $1.5 M_\odot$. Clearly, our knowledge of the equation of state is not accurate enough to distinguish between this value and those of equation (6).

Brown (1992) has shown that with the presence of the kaon condensate, hadronic matter has such a low energy density that it does not merge to quark matter until very high densities, $\sim 10\rho_0$ or greater.

We see, therefore, that the presence of a kaon condensate substantially softens the equation of state of dense matter and brings the maximum nucleon star mass $M_{\max}$ down substantially from the range $M_{\max} \lesssim 2 M_\odot$ generally given by equations of state without kaon condensate present.

Extensive calculations of higher order corrections to the kaon condensate in heavy fermion chiral perturbation theory (Brown et al. 1994) have confirmed the Politzer & Wise (1991) results that the kaon condensate is extremely robust. Thus, we believe in the result that the maximum nucleon star mass is substantially smaller, $M_{\max} \sim 1.5 M_\odot$, than obtained previously with conventional equations of state.

5. POSTSHOCK MECHANISM FOR FORMATION OF BLACK HOLES

Once the mass of the compact remnant exceeds $M_{\max}$, collapse into a black hole goes in a fraction of a millisecond. However, if this collapse happened in SN 1987A, it must have been more than 12 s after the first collapse into the compact object, because at 12 s a neutrino was still observed at Kamiokande II. The question is, then how the core can be stable for such a long period of nucleosynthesis and neutrino emission.

It has been suggested for some time that stars heavier than $\sim 25 M_\odot$ may leave black holes (e.g., Wilson et al. 1986), also that such stars first explode, exhibiting light curves of Type II supernovae and returning matter to the galaxy, and then collapse into black holes (Woosley & Weaver 1986). The compact core is, for a certain range of masses, stabilized by thermal pressure during the period of Kelvin-Helmholtz contraction long enough to carry out nucleosynthesis, going into a black hole after cooling and deleptonization. Calculations by Bombaci, Prakash, & Brown (1993) show that with their equation of state and additional $\sim 0.1 M_\odot$ over and above the $M_{\max}$, where $M_{\max}$ is the maximum mass of the cold deleptonized core, can be stabilized by thermal pressure sufficiently long to accomplish nucleosynthesis (and to emit neutrinos for the 12 s period in which they were observed in Kamiokande during SN 1987A). The additional $\sim 0.1 M_\odot$ is a "window" in the core mass, in which nucleosynthesis can take place, with the core later going into black hole.

In the "conventional scenario" (i.e., without K^- condensation) the window is only that produced by thermal pressure. One might think that the considerable pressure pro-

duced by trapped neutrinos in the explosion would stabilize some additional mass. However, just after the explosion Y_l is about 0.4, and therefore the nuclear symmetry energy is much smaller than for neutron matter, $Y_l \sim 0$. In fact, the lower symmetry energy for the higher Y_l decreases the pressure more than the leptons build it up. In the calculations of Bombaci et al. (1993) the "window" created by thermal pressure is decreased by $\sim 0.05 M_\odot$. This "window" is, therefore, relatively small in the conventional theory, although equations of state other than that used by Bombaci et al. might give a somewhat larger one.

With kaon condensation the situation is very different. The high-density matter remains nuclear matter, as we discussed in the last section, so that our $M_{\max} \sim 1.5 M_\odot$ applies to this nuclear matter star. The proton fraction $Y_p \sim 0.4$, which exists after explosion, remains the same as we go to nuclear matter with K^- condensation, so that no symmetry energy needs to be paid. Thus, as the leptons are emitted, the nuclear interaction energy remains roughly the same; we simply lose kinetic energy of the leptons. Consequently, the leptons can stabilize a substantially larger core than $M_{\max}$ for short times. This was recently pointed out to us by Lattimer & Prakash (1993) and we summarize their work here. They find that with the nuclear compression modulus of 190 MeV, which gives the cold, $Y_l \sim 0$ $M_{\max} \sim 1.5 M_\odot$, a mass of $1.7 M_\odot$ is stable for short times, because of the lepton pressure. Inclusion of the thermal pressure mentioned above will bring this mass up to $\sim 1.8 M_\odot$.

As the trapped neutrinos are emitted, bringing the lepton pressure down over a period of greater than 12 s, the lepton pressure decreases, but the nucleons in the core are heated because of entropization of the matter by the neutrino flux, analogous to Joule heating in a wire (Burrows & Lattimer 1986) and the thermal pressure is substantially increased, sufficiently so that the compact core continues to be stabilized until it cools.

The postshock accretion, discussed by Brown et al. (1992b) plays a lesser role. This accretion has recently been discussed by Bethe (1993), who had argued that in the supernova shock there is vigorous convection, but that at some time (2 s in SN 1987A) convection stops because heat is no longer supplied by neutrinos. When this happens, a substantial fraction of the previously convecting material falls into the neutron star at the center. He estimates this fraction to be about 10% of the mass in the shock wave, or about $0.04 M_\odot$. Most of this mass will accrete after the ~ 12 s it took for the neutrons to be emitted in SN 1987A.

The combination of stabilization of an additional $0.2 M_\odot$ by leptons for $t > 12$ s, $0.1 M_\odot$ by thermal pressure and the postshock accretion, means that compact cores which have late-time cold masses of up to $1.84 M_\odot$ will be stable during this ~ 12 s and possibly longer, even if the maximum stable cold mass is only $M_{\max} \sim 1.5 M_\odot$.

From these arguments and the baryon number masses of Timmes et al. (1993) which we discussed in § 3, we see that stars of mass up to $\sim 30 M_\odot$ will be able to become supernovae and thus return elements to the galaxy, but most of those in the range of 18 – $30 M_\odot$ will later go into low-mass black holes. Most stars heavier than $30 M_\odot$ will go directly into high-mass black holes.

We return to a discussion of Type Ib supernovae. Wheeler & Leverault (1985) came to the conclusion that the physics of Ib explosions was similar to that of Type II supernovae. Ensmann & Woosley (1988) found, through detailed light curve calcu-

lations, that the Type Ib progenitors were 4–7 $M_{\odot}$ He stars, corresponding to 15–25 $M_{\odot}$ mainsequence stars. These authors suggested that the Type Ib progenitors are Wolf-Rayet stars. Swartz & Wheeler (1991) suggest that it may be better to discern the mass of exploding Wolf-Rayet stars from their late-time light curve, rather than near maximum, as studied by Ensman & Woosley. Assuming the V light curve to be parallel to the bolometric light curve, Swartz & Wheeler conclude that the mass of the ejecta of the Type Ib SN 1984L must have been in excess of 10 $M_{\odot}$, and all of this in He or heavier elements. A He star of 10 $M_{\odot}$ corresponds to a main-sequence star of mass $\sim 30 M_{\odot}$. A similar argument can be made for SN 1985F. From our estimates, this should be possible, with the core SN 1984L having gone, following nucleosynthesis, into a black hole. On the other hand, stars somewhat heavier than $\sim 30 M_{\odot}$ will go directly into black holes, as suggested by Wheeler & Shields (1976) for the progenitor of the black hole of mass $\gtrsim 9 M_{\odot}$ in Cygnus X-1.

We are led to conclude that the cutoff for the return of elements to the Galaxy should be at $\sim 30 M_{\odot}$, at the upper end of the estimate, equation (3). This is in agreement with the estimate of S. E. Woosley (private communication). As noted in § 3, the compact core masses do not, however, increase monotonically with stellar mass. Thus, some stars of mass greater than 30 $M_{\odot}$ may explode, returning matter to the Galaxy, whereas others of mass greater than 30 $M_{\odot}$ may go directly into black holes.

6. IMPLICATIONS OF OUR SCENARIO FOR BLACK HOLE FORMATION

Properties of the best candidates for black holes; e.g., Cyg X-1, LMC X-3, A0620–00, and GS 2023+338, have been reviewed recently by van den Heuvel (1992). These are relatively heavy, the most likely masses falling in the range of 9–16 $M_{\odot}$. In our scenario, such heavy mass black holes would be formed by the nearly instantaneous collapse, without nucleosynthesis, of large stars $M \gtrsim 30 M_{\odot}$.

Lighter black holes with mass $M_{\text{BH}} \simeq 1.5 M_{\odot}$ should result from the collapse of most of the stars in mass range $18 M_{\odot} \lesssim M \lesssim 30 M_{\odot}$, according to our scenario. These events are

accompanied by nucleosynthesis. If they occur on binaries, then the binary may be disrupted in the explosion.

Taking the present supernova rate in late-type spiral galaxies like our own to be $\sim 0.03 \text{ yr}^{-1}$ and correcting for a higher rate for the first few billion years after the formation of the Galaxy, van den Heuvel (1992) estimates that there are at least some 10^9 neutron stars in the Galaxy. Whereas van den Heuvel estimates the ratio f of neutron stars to black holes in the Galaxy to be

$$10 < f < 45, \quad (7)$$

we find roughly equal numbers of neutron stars and black holes; i.e., $f \sim 1$. Thus, we expect about 10^9 black holes in the Galaxy.

Note added in manuscript.—After completion of our manuscript, we received a paper by N. Pratzos to be published in A&A in which he shows that observations of the carbon/oxygen ratio in halo stars require all stars up to ~ 90 – $100 M_{\odot}$ to contribute to galactic nucleosynthesis. Uncertainties in the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ rate do, however, affect this argument. N. Pratzos (private communication) stressed also that some oxygen from SN II's, locked up in grains, may have escaped the shallow potential of dwarf galaxies where $\Delta Y/\Delta Z$ was measured, changing the value of this quantity. Whereas Pagel et al. (1992) did correct for this loss, some uncertainty remains. In view of Pratzos's arguments, we believe that the arguments of Maeder, although extremely interesting, may not be as firm as we indicated in the text.

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